Lecture - 11

Chapter 5

IMAGE RESTORATION AND RECONSTRUCTION

Image Degradation/Restoration Model

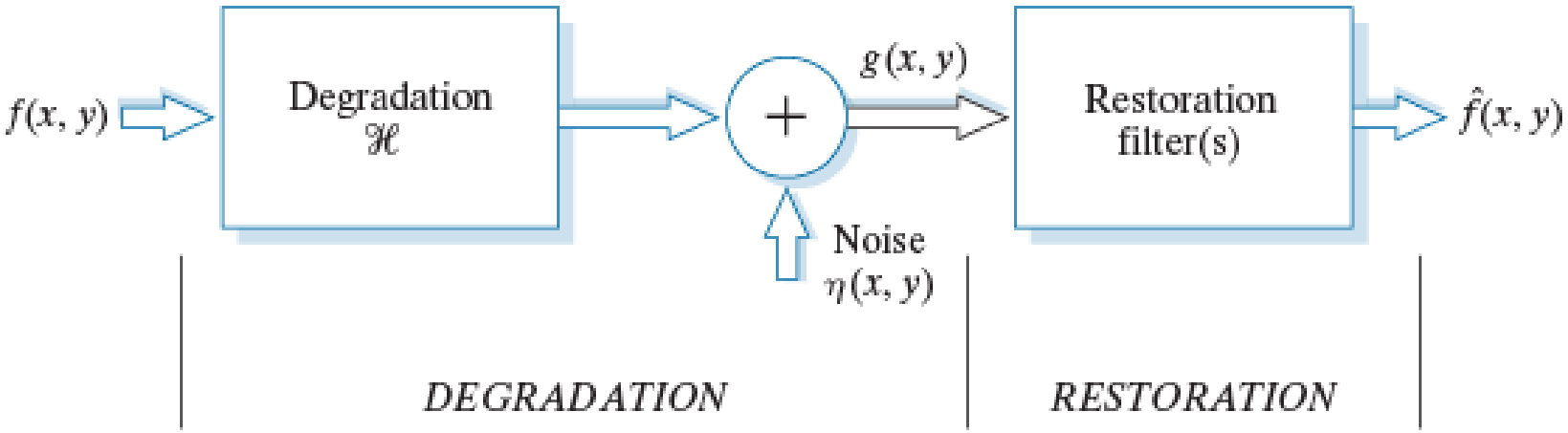
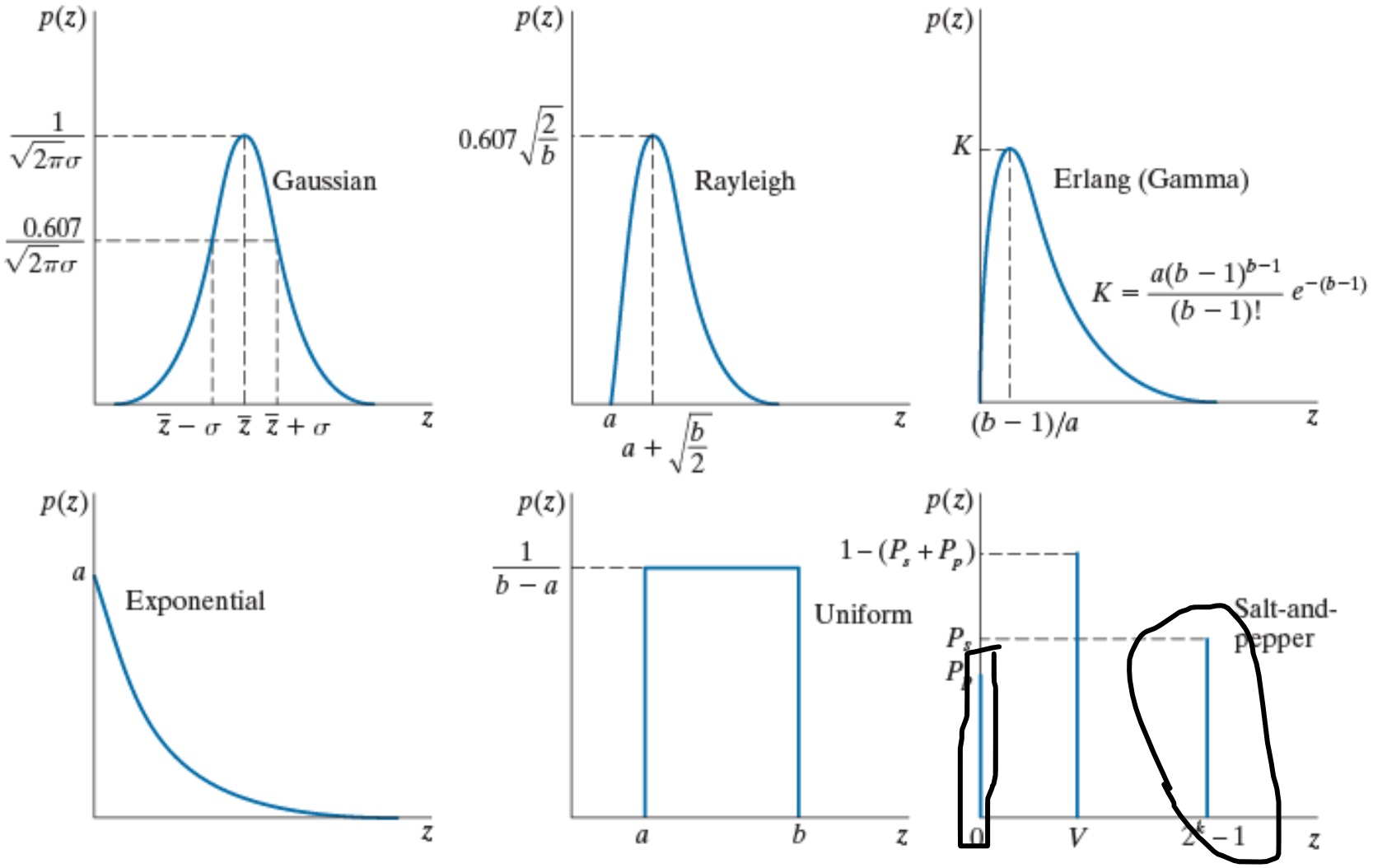


FIGURE 5.1 A model of the image degradation/restoration process.

Noise Models



a b c
d e f

FIGURE 5.2 Some important probability density functions.

Test Pattern

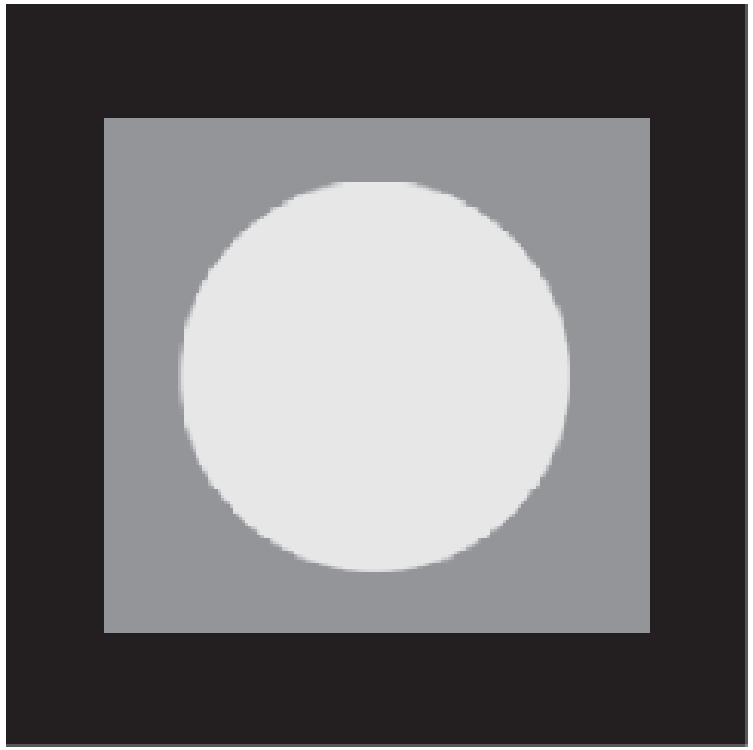
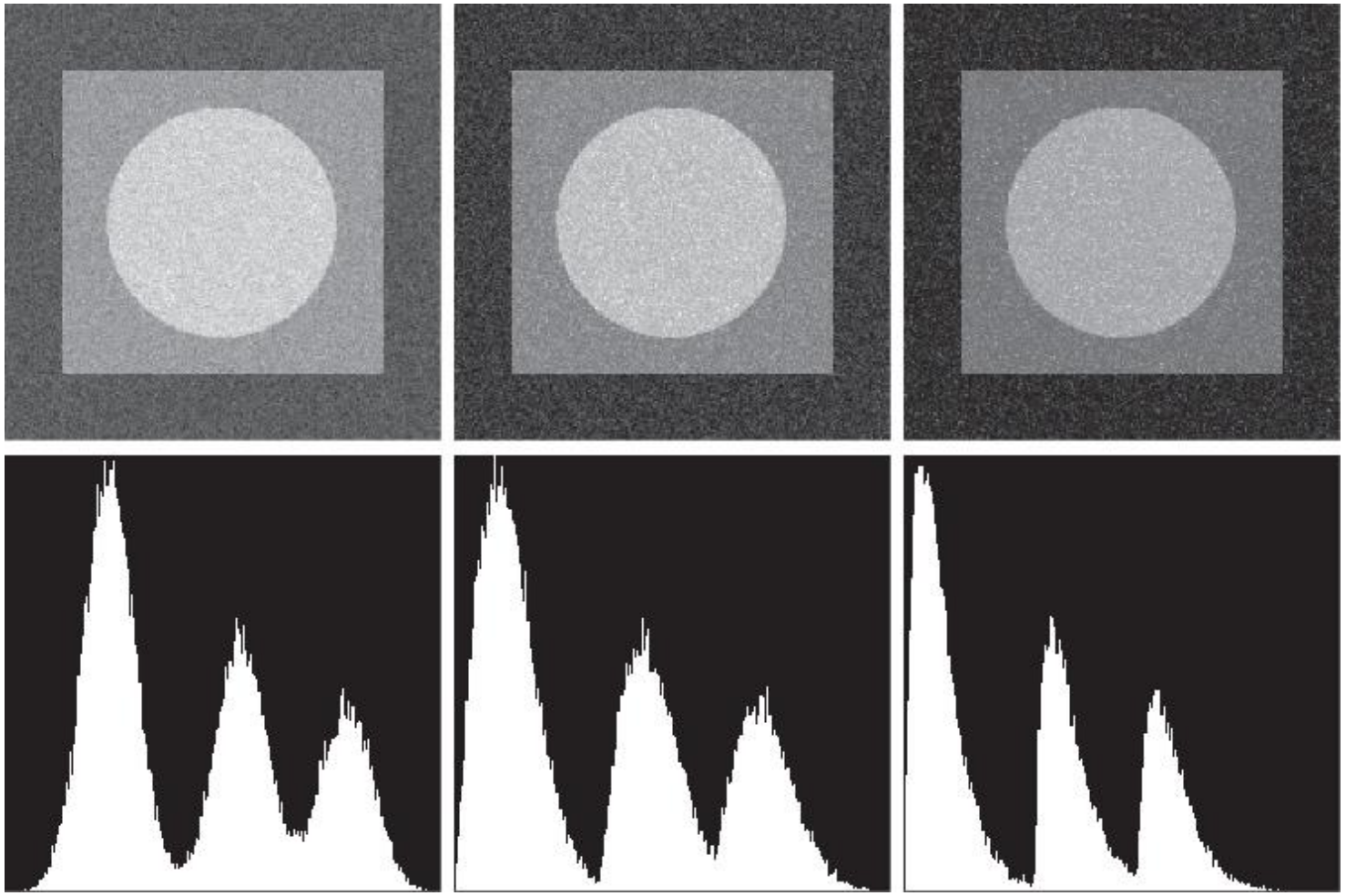


FIGURE 5.3
Test pattern used to illustrate the characteristics of the PDFs from Fig. 5.2.

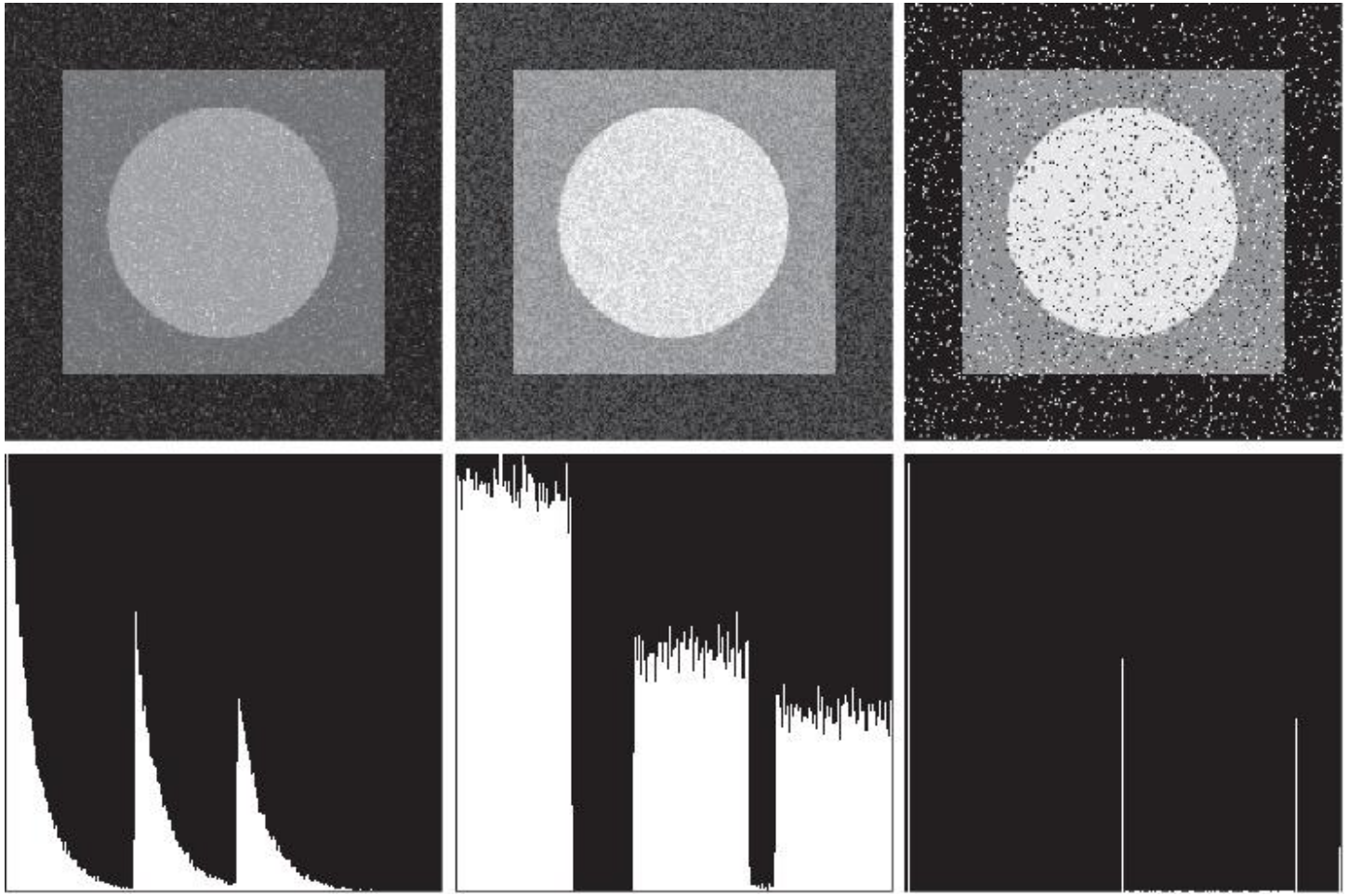
Test Pattern after Adding Noise



a b c
d e f

FIGURE 5.4 Images and histograms resulting from adding Gaussian, Rayleigh, and Erlanga noise to the image in Fig. 5.3.

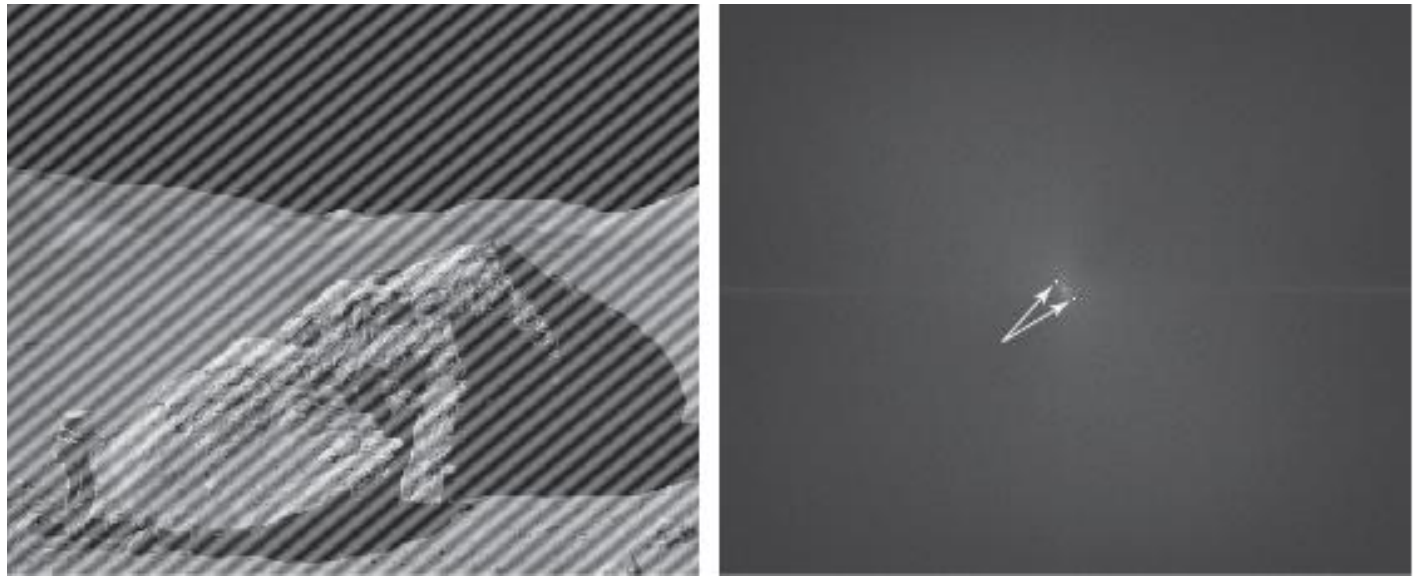
Test Pattern after Adding Noise



g h i
j k l

FIGURE 5.4 (*continued*) Images and histograms resulting from adding exponential, uniform, and salt-and-pepper noise to the image in Fig. 5.3. In the salt-and-pepper histogram, the peaks in the origin (zero intensity) and at the far end of the scale are shown displaced slightly so that they do not blend with the page background.

Sinusoidal Noise



a b

FIGURE 5.5 (a) Image corrupted by additive sinusoidal noise. (b) Spectrum showing two conjugate impulses caused by the sine wave. (Original image courtesy of NASA.)

Noisy Image Strip Histogram

a b c

FIGURE 5.6 Histograms computed using small strips (shown as inserts) from (a) the Gaussian, (b) the Rayleigh, and (c) the uniform noisy images in Fig. 5.4.

Restoration in the Presence of Noise Only (De-Noising)

$$g(x,y) = f(x,y) + \eta(x,y)$$

$$G(u,v) = F(u,v) + N(u,v)$$

mean = avg

mid

blur↑

- Mean filters
  - Arithmetic mean filter
    - $g(x,y)$ is the corrupted image
    - $S_{x,y}$ is the mask

$$\hat{f}(x,y) = \frac{1}{mn} \sum_{(s,t)\in S_{x,y}} g(s,t)$$

blur↓

  - Geometric mean filters
    - Tends to preserve more details

$$\hat{f}(x,y) = \left[\prod_{(s,t)\in S_{x,y}} g(s,t) \right]^{\frac{1}{mn}}$$

  - Harmonic mean filter
    - Works well for salt noise but fails for pepper noise

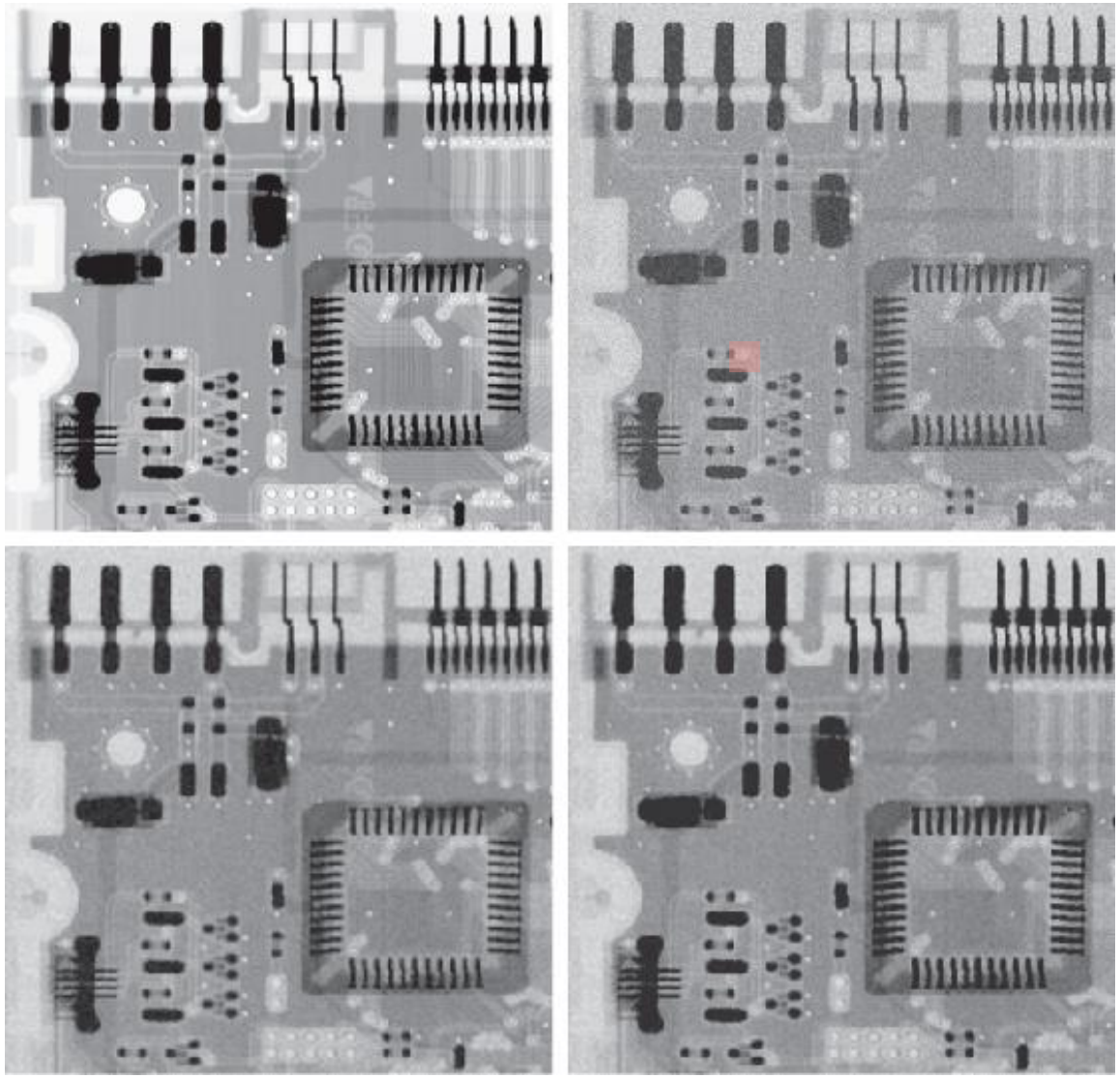
$$\hat{f}(x,y) = \frac{mn}{\sum_{(s,t)\in S_{x,y}} \frac{1}{g(s,t)}}$$

  - Contraharmonic mean filter
    - Q: order of the filter
    - Positive Q works for pepper noise
    - Negative Q works for salt noise
    - Q=0➔arithmetic mean filter
    - Q=-1➔harmonic mean filter

$$\hat{f}(x,y) = \frac{\sum_{(s,t)\in S_{x,y}} g(s,t)^{Q+1}}{\sum_{(s,t)\in S_{x,y}} g(s,t)^{Q}}$$

De-Noising

Corrupted by Gaussian Noise



a b
c d

FIGURE 5.7 (a) X-ray image of circuit board. (b) Image corrupted by additive Gaussian noise. (c) Result of filtering with an arithmetic mean filter of size 3×3. (d) Result of filtering with a geometric mean filter of the same size. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)

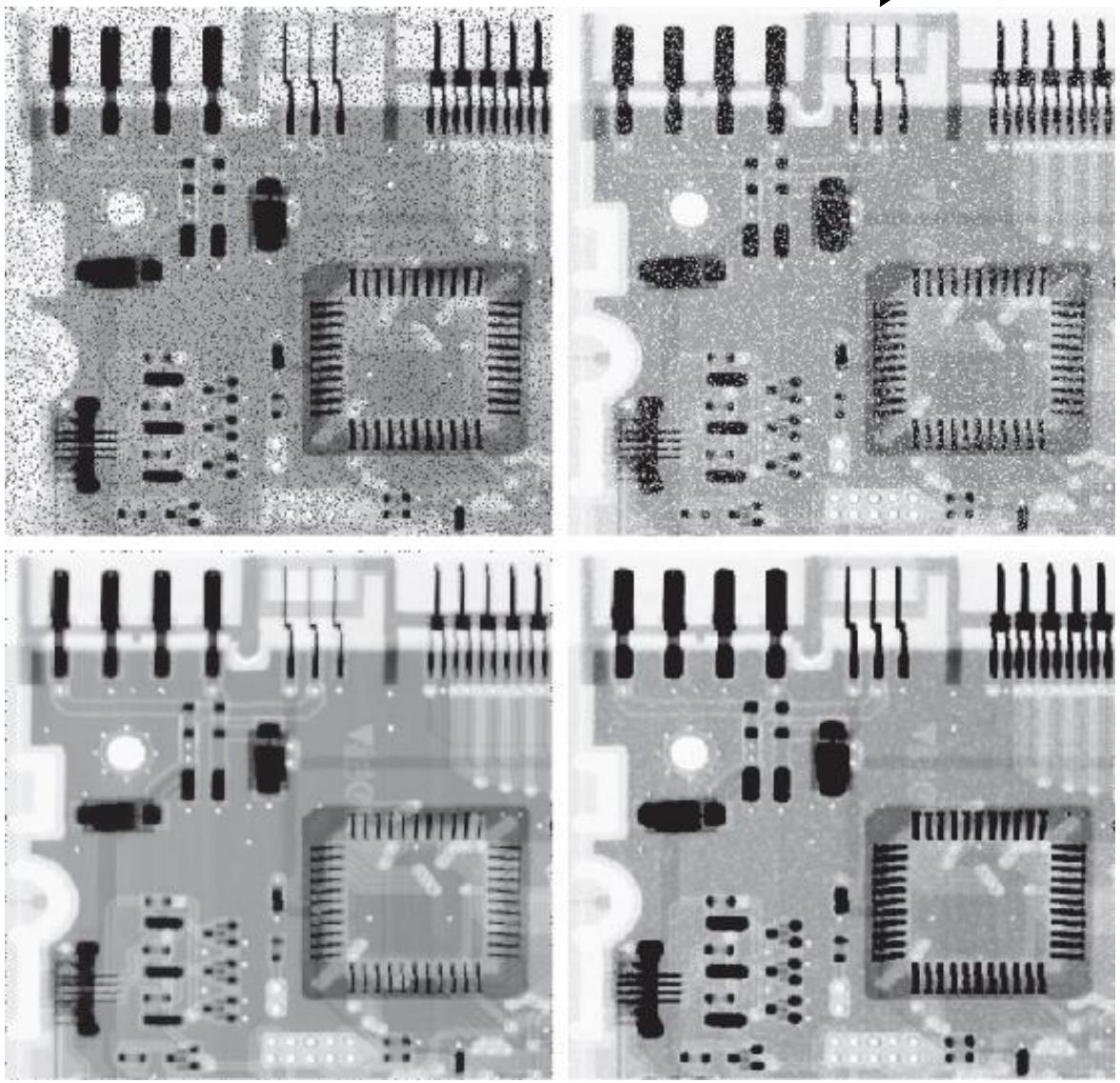
Arithmetic Mean Filtering

Geometric Mean Filtering

De-Noising

Corrupted by pepper noise

Corrupted by salt noise



a b
c d

FIGURE 5.8 (a) Image corrupted by pepper noise with a probability of 0.1. (b) Image corrupted by salt noise with the same probability. (c) Result of filtering (a) with a 3×3 contra-harmonic filter $Q = 1.5$. (d) Result of filtering (b) with $Q = -1.5$.

+ → B

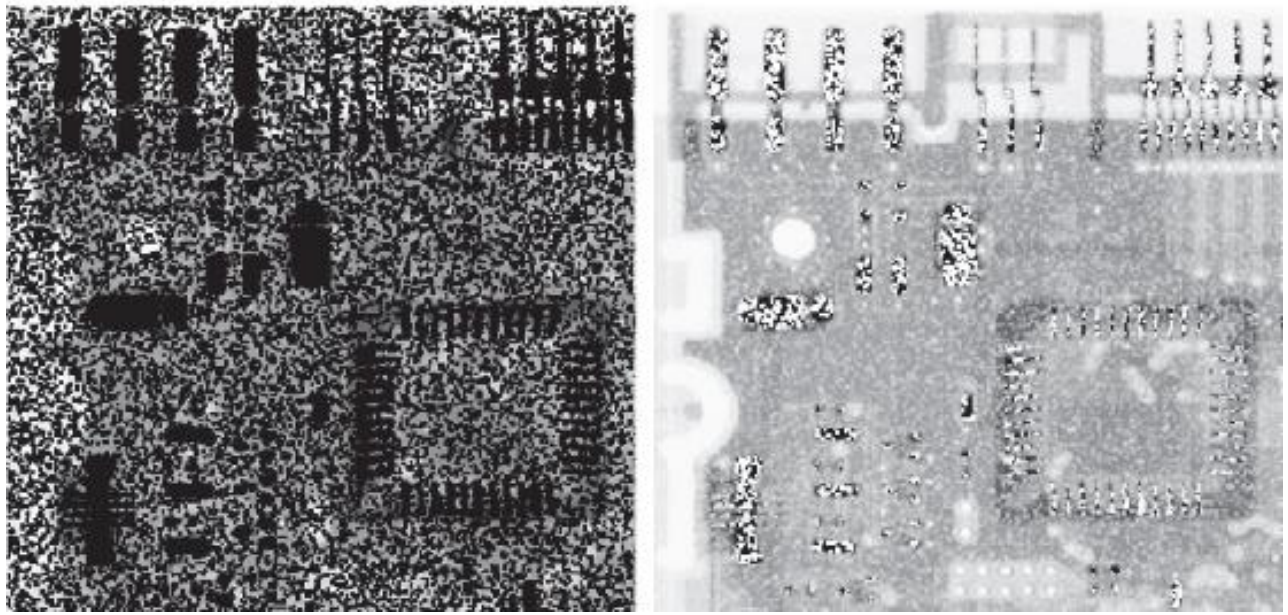
− → W

3x3 Contraharmonic Q=1.5

3x3 Contraharmonic Q=-1.5

De-Noising

wrong choose



a b

FIGURE 5.9 Results of selecting the wrong sign in contraharmonic filtering. (a) Result of filtering Fig. 5.8(a) with a contraharmonic filter of size 3×3 and $Q = -1.5$. (b) Result of filtering Fig. 5.8(b) using $Q = 1.5$.

3x3
Contraharmonic
Q=-1.5

3x3
Contraharmonic
Q=1.5

Filters based on Order Statistics (De-Noising)

sorted Sort > pick center

- Median filter
  - Median represents the 50th percentile of a ranked set of numbers

$$\hat{f}(x,y) = \underset{(s,t)\in S_{x,y}}{\text{median}}\{g(s,t)\}$$

- Max and Min filter
  - Max filter uses the 100th percentile of a ranked set of numbers
    - Good for removing pepper noise → swap by Max (bright)
  - Min filter uses the 1 percentile of a ranked set of numbers
    - Good for removing salt noise

sorted

- Midpoint filter
  - Works best for noise with symmetric PDF like Gaussian or uniform noise

$$\hat{f}(x,y) = \frac{1}{2}\left[\max_{(s,t)\in S_{xy}}\{g(s,t)\} + \min_{(s,t)\in S_{xy}}\{g(s,t)\}\right]$$

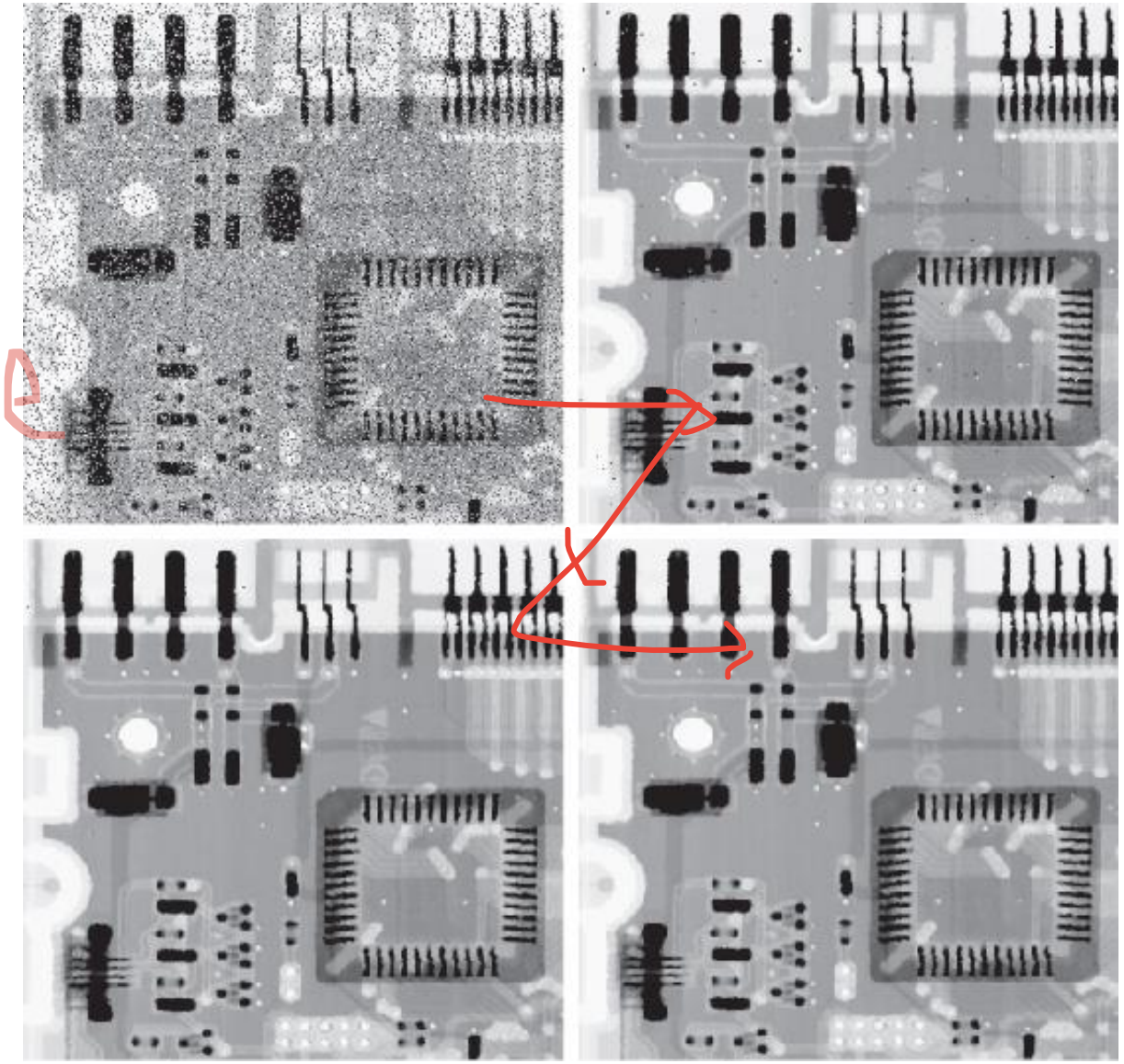
De-Noising

Corrupted by salt & pepper noise

One pass median filtering

not blurring

more Pass ↓ noise



a b
c d

FIGURE 5.10 (a) Image corrupted by salt-and- pepper noise with probabilities $P_s = P_p = 0.1$. (b) Result of one pass with a median filter of size 3×3. (c) Result of processing (b) with this filter. (d) Result of processing (c) with the same filter.

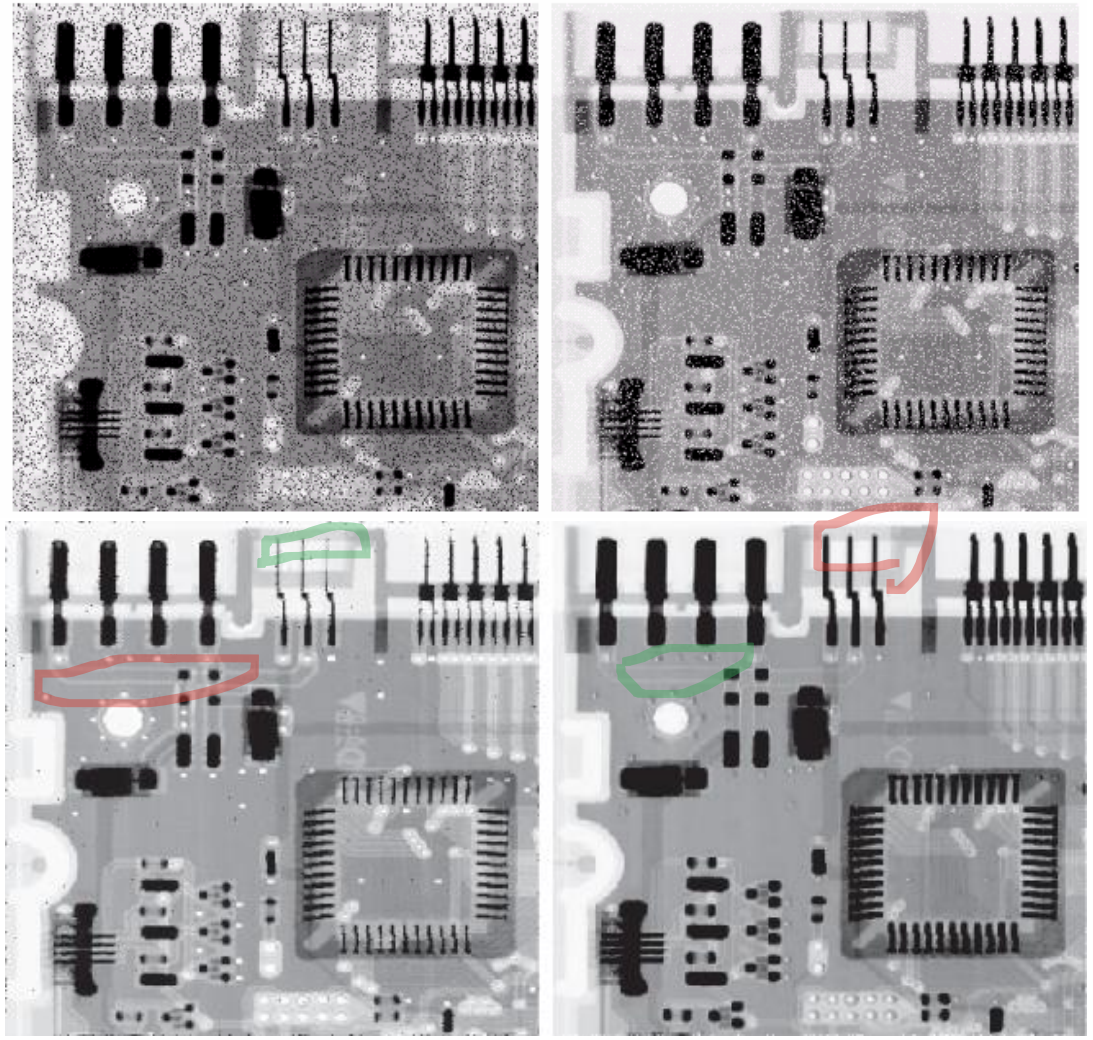
Two pass median filtering

Three pass median filtering

De-Noising

Corrupted by pepper noise

Corrupted by Salt noise



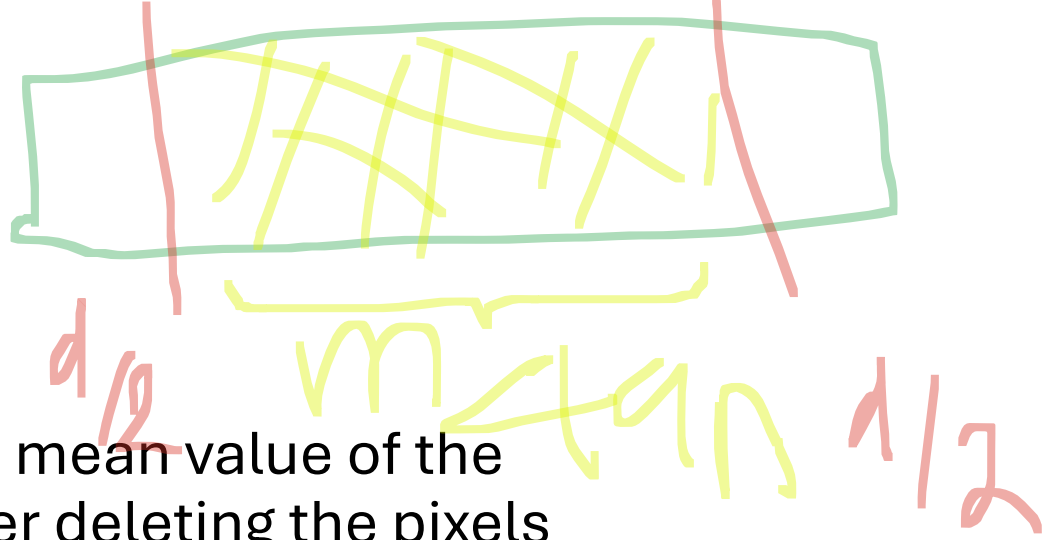
a b

FIGURE 5.11 (a) Result of filtering Fig. 5.8(a) with a max filter of size 3×3. (b) Result of filtering Fig. 5.8(b) with a min filter of the same size.

Max Filtering

Min Filtering

Alpha-Trimmed Mean Filter (De-Noising)



- Alpha-trimmed mean filter takes the mean value of the pixels enclosed by an $m \times n$ mask after deleting the pixels with the $d/2$ lowest and the $d/2$ highest gray-level values

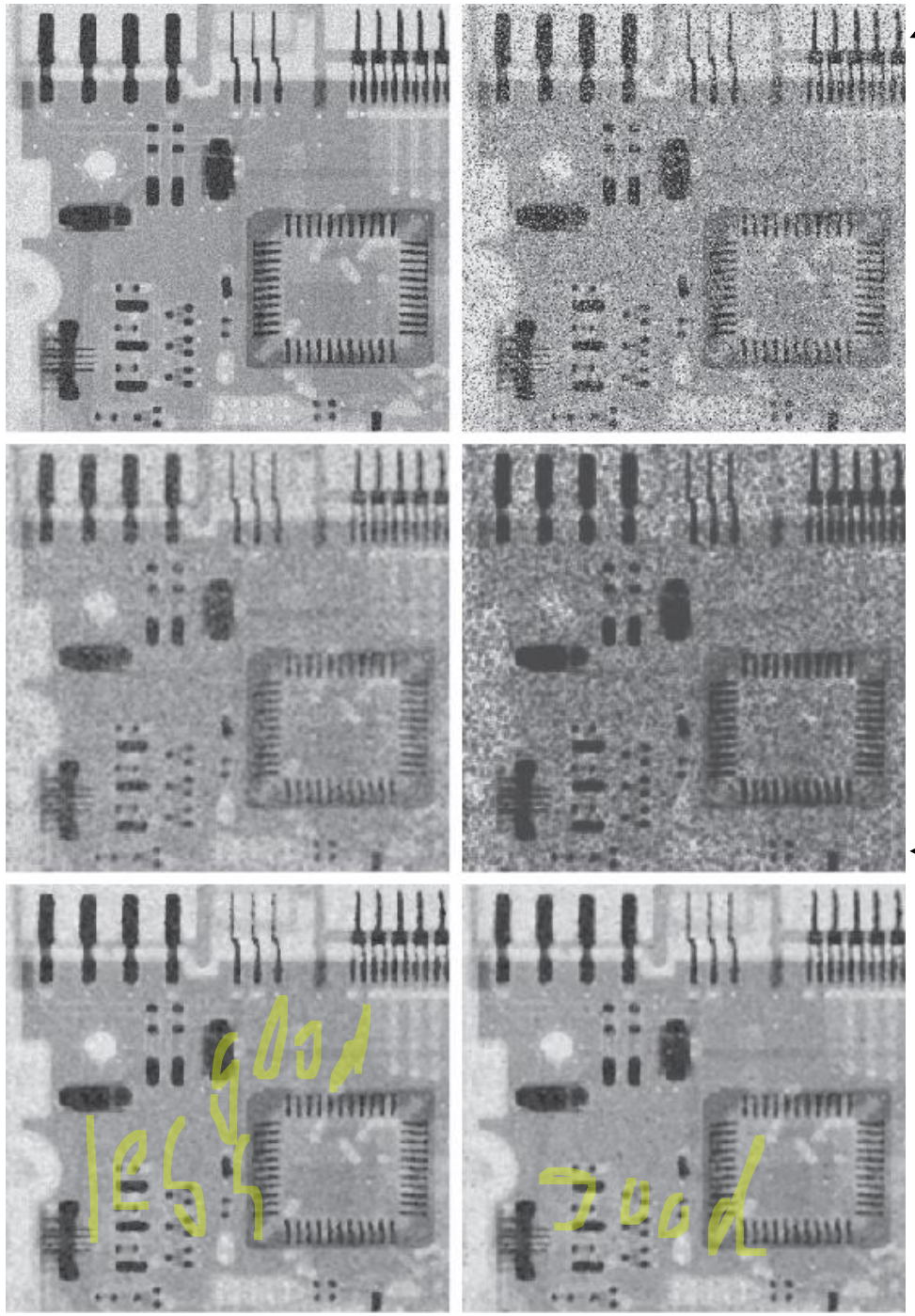
$$\hat{f}(x,y) = \frac{1}{\boxed{mn} - d} \sum_{(s,t)\in S_{xy}} g_r(s,t)$$

  - $g_r(s,t)$ represent the remaining mn-d pixels
  - It is useful in situations involving multiple types of noise like a combination of salt-and-pepper and Gaussian

De-Noising

Corrupted by Additive Uniform Noise

Added Salt & Pepper noise



a b
c d
e f

FIGURE 5.12 (a) Image corrupted by additive uniform noise. (b) Image additionally corrupted by additive salt-and-pepper noise. (c)-(f) Image (b) filtered with a 5×5: (c) arithmetic mean filter; (d) geometric mean filter; (e) median filter; (f) alpha-trimmed mean filter, with $d = 6$.

5x5 Arithmetic Mean Filtering

5x5 Geo-Mean Filtering

5x5 Median Filtering

5x5 Alpha-trimmed Mean Filter

Adaptive Filters (De-Noising)

- Adaptive Local Noise Reduction Filter
  - Assume the variance of the noise σ_η^2 is either known or can be estimated satisfactorily
  - Filtering operation changes at different regions of an image according to local variance σ_L^2 calculated within an M×N region
  - If $\sigma_L^2 > \sigma_\eta^2$, the filtering operation is defined as

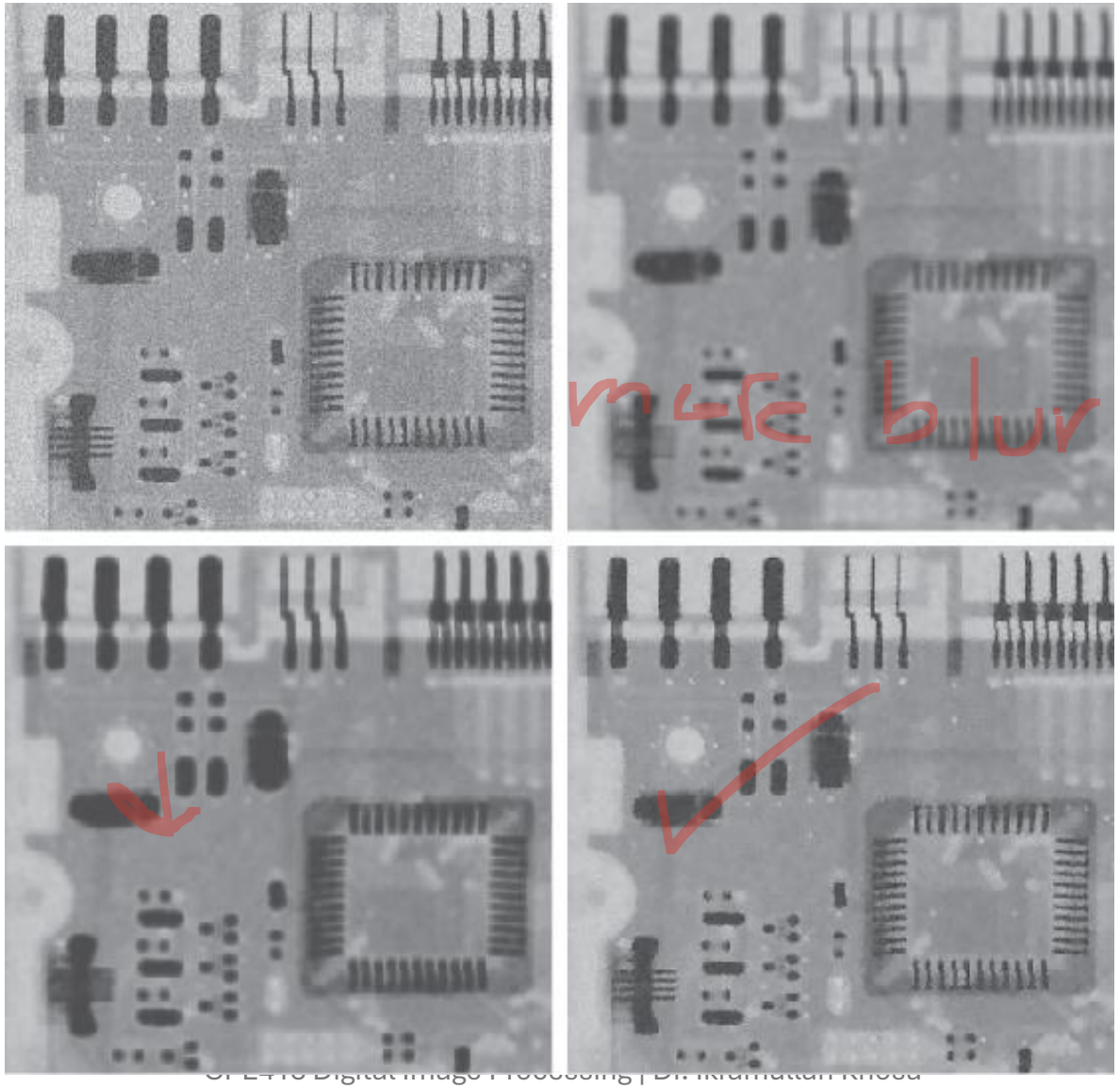
$$\hat{f}(x,y) = g(x,y) - \frac{\sigma_\eta^2}{\sigma_L^2}[g(x,y) - m_L]$$

  - If $\sigma_L^2 < \sigma_\eta^2$, the output takes the mean value
    - That is: $\frac{\sigma_\eta^2}{\sigma_L^2}$ is set to be 1
  - At edges, it is assumes that $\sigma_L^2 > \sigma_\eta^2$

De-Noising

Corrupted by Gaussian noise

Arithmetic Mean Filtering



a b
c d

FIGURE 5.13 (a) Image corrupted by additive Gaussian noise of zero mean and a variance of 1000. (b) Result of arithmetic mean filtering. (c) Result of geometric mean filtering. (d) Result of adaptive noise-reduction filtering. All filters used were of size 7×7.

Geo-Mean Filtering

Adaptive Filtering

Adaptive Median Filter (De-Noising)

- Median filter is effective for removing salt-and-pepper noise
  - The density of the impulse noise can not be too large
- Adaptive median filter
  - Notation
    - Z_{min}: minimum gray value in S_{xy}
    - Z_{max}: maximum gray value in S_{xy}
    - Z_{med}: median of gray levels in S_{xy}
    - Z_{xy}: gray value of the image at (x,y)
    - S_{max}: maximum allowed size of S_{xy}

Adaptive Median Filter (De-Noising)

- Two levels of operations
  - Level A:
    - $A1 = Z_{med} - Z_{min}$
    - $A2 = Z_{med} - Z_{max}$
    - If $A1 > 0$ AND $A2 < 0$, Go to level B else increase the window size by 2
    - If window size <= S_{max} repeat level A else output Z_{xy}

Used to test whether Z_{med} is part of s-and-p noise. If yes, window size is increased

  - Level B:
    - $B1 = Z_{xy} - Z_{min}$
    - $B2 = Z_{xy} - Z_{max}$
    - If $B1 > 0$ AND $B2 < 0$, output Z_{xy} else output Z_{med}

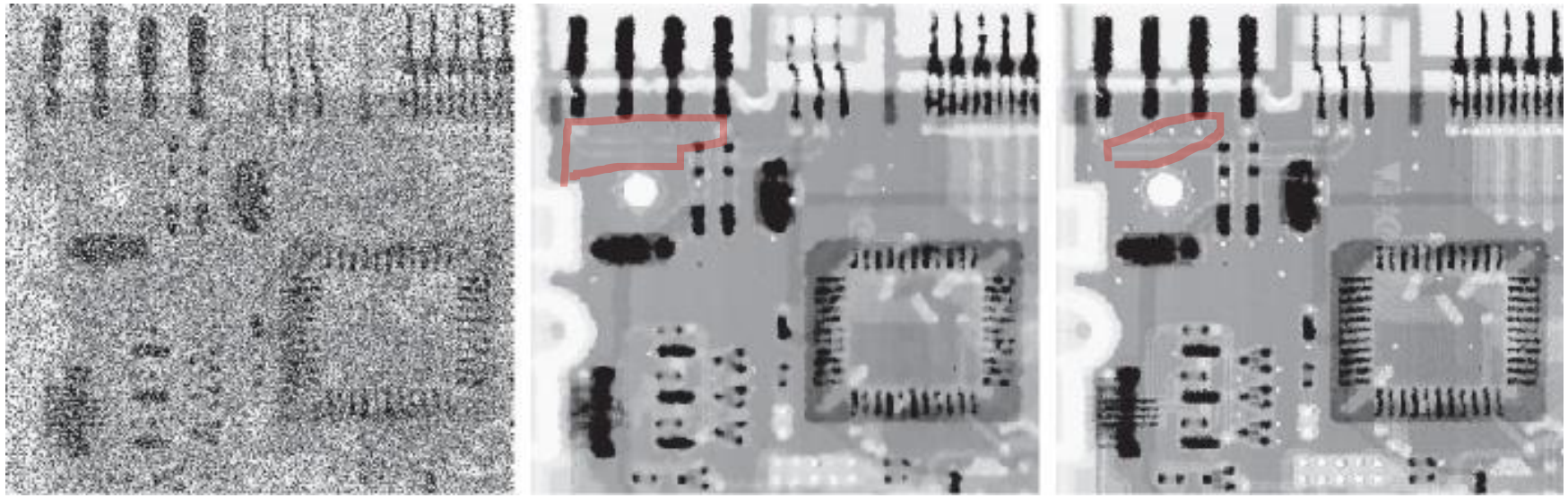
Used to test whether Z_{xy} is part of s-and-p noise. If yes, apply regular median filtering

De-Noising

original content replace.

Median Filtering

Adaptive Median Filtering



a b c

FIGURE 5.14 (a) Image corrupted by salt-and-pepper noise with probabilities $P_s = P_p = 0.25$. (b) Result of filtering with a 7×7 median filter. (c) Result of adaptive median filtering with $S_{max} = 7$.